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A. Project Overview

- Goal: What question are you answering?

Project_Overview: Analyze the changes in gender plus racial/ethnic groups from the data given have experienced the most significant rise in stimulant-related overdose deaths since 2010?

- Dataset: Source, size (link if too large for GitHub).

Link:

<https://catalog.data.gov/dataset/drug-overdose-death-rates-by-drug-type-sex-age-race-and-hispanic-origin-united-states-3f72f>

Has 6229 rows and 15 columns

B. Data Processing

- How you loaded it into Rust.

I loaded the data into Rust by making a folder called data/ and then just importing the CSV file. Then load `csv::ReaderBuilder` in `csv_reader.rs` so I could put into structs and read the CSV with headers.

- Any cleaning or transformations applied.
 - I did row filtering which skipped any record missing a year, drug label, demographic or rate because we wouldn't know how to take them into account when doing the math.
 - Also labeling normalization edited the named of one of the forms of drugs to include - "unnamed" because if not it just said "All drug overdose deaths".
 - Parsing so I converted year to u16, estimate to f64, retained panel and stub label as string.
 - Finally had a demographic filter in `main.rs` this was to keep only `race_ethnicity` entries starting with male or female and drop any that have numbers

C. Code Structure

1. Modules

- `Data_model`: here we define `overdoserecord` hihc is the core type for each csv row, and `Analysis` as enum, more information in the code
- `Csv_reader`: here we read the raw CSV and then it returns `Vec<OverdoseRecord>`, and handled CSV parsing and normalizaation, more information in the code
- `Analysis`: we have function that filter and the change in rate algorithms, like `rate_by_race_in_year` that maps each demographic to its rate in a given year, more information in the code
- `Main.rs`: Here we orchestrate the flow by printing the results, filters male/female demographics, have the analyses, uses clusters abovebelow avg, etc

2. Key Functions & Types (Structs, Enums, Traits, etc)

- `read_records(path: &str) -> Result<Vec<OverdoseRecord>>`

The purpose of this is to load and clean the raw CSV file and the it returns a list of valid records when you put a file path. The whole point is to locate headers, iterates row,parses fields, skip blank rows, and normalizes labels.

- `OverdoseRecord` (struct)

The purpose of this struct is so have a strongly typed container for year, drug type, race ethnicity and rate. What you get in return is constructed from the four parsed CSV fields. The whole point of this struct is just to old data.

- `run_analysis_and_plot(label: &str, subset: &[OverdoseRecord]) -> Result<(),Error>`

The codes only purpose is to print the change of rate sorted and show its top performer and cluster above and below avg.

- `filter_by_keyword(records: &[...], kw: &str) -> Vec<OverdoseRecord>`

It select records whose drug type which will be contained case insensitive. Its just a oneliner iterator chain.

- `increase_since_2010(records: &[...]) -> HashMap<String, f64>`

Here for each demographic we compute change of rate from 2018-2010 and by grouping the race ethnicity, we find the 2010 baseline via `find()` and insert differences.

3. Main Workflow

- Firstly we will load all records using `csv_reader::read_records`, then we will filter between the two genders in the chart which are female and male, then enumerate

each unique drug type in the CSV file, so for each drug we will call the function `run_analysis_and_plot` to print a table and the stats. Finally run a combined analysis on all drugs with the same table and stats as the other.

D. Tests

- cargo test output (paste logs or provide screenshots).

```
running 2 tests
test tests::test_filter_by_keyword ... ok
test tests::test_increase_since_2010 ... ok
test result: ok. 2 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s

Running unittests src/main.rs (target/debug/deps/rust_project-4f28d39c5818a168)

running 0 tests
test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s

Doc-tests rust_project

running 0 tests
test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s
@Angier18 → /workspaces/rust-project/rust-project (main) $ █
```

- For each test: what it checks and why it matters.
 - So I am checking that `test_filter_by_keyword` is returning, case insensitive, keywords, the ones we are looking for, because we want to know that each drug is added into the stats for a better representation of the data.
 - And for `test_increase_since_2010` it will check that the change in rate is done correctly when the 2010 baseline is there and that the demographics without 2010 record are not included in the data because this is what our data will be outputting and if it's wrong then our data is wrong.

F. Usage Instructions

- How to build and run your code.
 - Build:

`cd rust-project`

`cargo build`

- Run:

```
cargo run --  
data/Drug_overdose_death_rates__by_drug_type__sex__age__race__and_Hispanic_origin__Uni  
ted_States.csv
```

- Description of any command-line arguments or user interaction in the terminal.

My main.rs doesn't parse any flags or parameters it just expects the CSV at the location in data.

- Include expected runtime especially if your project takes a long time to run.

When running cargo build target(s) in 1m 03s

When running run target(s) in 0.18s

E. Results

- All program outputs (screenshots or pasted).

==== Analysis for: All drug overdose deaths: unnamed drug ====

Race/Ethnicity	Change in rate since 2010
----------------	---------------------------

Male: Not Hispanic or Latino: Black	+21.90
Male: Black or African American	+21.40
Male: Not Hispanic or Latino: White	+14.80
Male: Not Hispanic or Latino: American Indian or Alaska Native	+14.10
Male: White	+13.30
Male: Hispanic or Latino: All races	+9.00
Male: American Indian or Alaska Native	+8.50
Female: Not Hispanic or Latino: American Indian or Alaska Native	+5.90
Female: Not Hispanic or Latino: Black	+5.70
Female: Not Hispanic or Latino: White	+5.50
Female: Black or African American	+5.50
Female: White	+4.40
Female: American Indian or Alaska Native	+3.40
Male: Asian or Pacific Islander	+3.20
Male: Not Hispanic or Latino: Asian or Pacific Islander	+3.00
Female: Hispanic or Latino: All races	+1.60
Female: Not Hispanic or Latino: Asian or Pacific Islander	+0.70
Female: Asian or Pacific Islander	+0.60

Highest change in rate: Male: Not Hispanic or Latino: Black at +21.90

Average change in rate: 7.92

Above avg (7):

- Male: Black or African American +21.40
- Male: American Indian or Alaska Native +8.50
- Male: Not Hispanic or Latino: White +14.80
- Male: White +13.30
- Male: Not Hispanic or Latino: American Indian or Alaska Native +14.10
- Male: Not Hispanic or Latino: Black +21.90
- Male: Hispanic or Latino: All races +9.00

Below avg (11):

- Female: Not Hispanic or Latino: White +5.50
- Female: White +4.40
- Female: Hispanic or Latino: All races +1.60
- Male: Not Hispanic or Latino: Asian or Pacific Islander +3.00
- Female: Not Hispanic or Latino: Asian or Pacific Islander +0.70
- Female: Not Hispanic or Latino: Black +5.70
- Female: Not Hispanic or Latino: American Indian or Alaska Native +5.90
- Female: American Indian or Alaska Native +3.40
- Female: Asian or Pacific Islander +0.60
- Female: Black or African American +5.50
- Male: Asian or Pacific Islander +3.20

==== Analysis for: Drug overdose deaths involving any opioid ====

Race/Ethnicity	Change in rate since 2010
----------------	---------------------------

Male: Not Hispanic or Latino: Black	+17.20
Male: Black or African American	+16.80
Male: Not Hispanic or Latino: White	+13.70
Male: White	+12.20
Male: Hispanic or Latino: All races	+7.50
Male: Not Hispanic or Latino: American Indian or Alaska Native	+6.80
Female: Not Hispanic or Latino: White	+5.70
Female: Not Hispanic or Latino: Black	+5.00
Female: Black or African American	+4.90
Female: White	+4.70
Male: American Indian or Alaska Native	+4.30
Female: Not Hispanic or Latino: American Indian or Alaska Native	+2.10
Female: Hispanic or Latino: All races	+1.70
Male: Asian or Pacific Islander	+1.60
Male: Not Hispanic or Latino: Asian or Pacific Islander	+1.50
Female: American Indian or Alaska Native	+1.20
Female: Asian or Pacific Islander	+0.30
Female: Not Hispanic or Latino: Asian or Pacific Islander	+0.30

Highest change in rate: Male: Not Hispanic or Latino: Black at +17.20

Average change in rate: 5.97

Above avg (6):

- Male: Hispanic or Latino: All races +7.50
- Male: Black or African American +16.80
- Male: Not Hispanic or Latino: White +13.70
- Male: Not Hispanic or Latino: Black +17.20
- Male: White +12.20
- Male: Not Hispanic or Latino: American Indian or Alaska Native +6.80

Below avg (12):

- Female: Hispanic or Latino: All races +1.70
- Female: White +4.70
- Female: Not Hispanic or Latino: Black +5.00
- Male: Asian or Pacific Islander +1.60
- Male: American Indian or Alaska Native +4.30
- Female: Not Hispanic or Latino: White +5.70
- Female: Black or African American +4.90
- Female: Asian or Pacific Islander +0.30
- Male: Not Hispanic or Latino: Asian or Pacific Islander +1.50
- Female: Not Hispanic or Latino: Asian or Pacific Islander +0.30
- Female: American Indian or Alaska Native +1.20
- Female: Not Hispanic or Latino: American Indian or Alaska Native +2.10

==== Analysis for: Drug overdose deaths involving heroin ====

Race/Ethnicity	Change in rate since 2010
----------------	---------------------------

Male: Not Hispanic or Latino: White	+6.80
Male: Not Hispanic or Latino: Black	+6.60
Male: Black or African American	+6.60
Male: White	+6.00
Male: Hispanic or Latino: All races	+3.40
Female: Not Hispanic or Latino: White	+2.50
Female: White	+2.20
Female: Not Hispanic or Latino: Black	+2.10
Female: Black or African American	+2.00
Female: Hispanic or Latino: All races	+0.90

Highest change in rate: Male: Not Hispanic or Latino: White at +6.80

Average change in rate: 3.91

Above avg (4):

- Male: White	+6.00
- Male: Black or African American	+6.60
- Male: Not Hispanic or Latino: White	+6.80
- Male: Not Hispanic or Latino: Black	+6.60

Below avg (6):

- Female: Not Hispanic or Latino: White	+2.50
- Female: Not Hispanic or Latino: Black	+2.10
- Female: White	+2.20
- Female: Hispanic or Latino: All races	+0.90
- Female: Black or African American	+2.00
- Male: Hispanic or Latino: All races	+3.40

==== Analysis for: Drug overdose deaths involving methadone ====

Race/Ethnicity	Change in rate since 2010
----------------	---------------------------

Male: Not Hispanic or Latino: Black	+0.30
Male: Black or African American	+0.30
Female: Black or African American	+0.10
Female: Not Hispanic or Latino: Black	+0.00
Female: Hispanic or Latino: All races	+0.00
Male: Hispanic or Latino: All races	-0.20
Female: White	-0.30
Female: Not Hispanic or Latino: White	-0.40
Male: White	-1.10
Male: Not Hispanic or Latino: White	-1.20

Highest change in rate: Male: Not Hispanic or Latino: Black at +0.30

Average change in rate: -0.25

Above avg (6):

- Female: Not Hispanic or Latino: Black	+0.00
- Male: Hispanic or Latino: All races	-0.20
- Male: Not Hispanic or Latino: Black	+0.30
- Female: Black or African American	+0.10
- Female: Hispanic or Latino: All races	+0.00
- Male: Black or African American	+0.30

Below avg (4):

- Male: White	-1.10
- Male: Not Hispanic or Latino: White	-1.20
- Female: Not Hispanic or Latino: White	-0.40
- Female: White	-0.30

==== Analysis for: Drug overdose deaths involving natural and semisynthetic opioids ====

Race/Ethnicity	Change in rate since 2010
----------------	---------------------------

Male: Not Hispanic or Latino: Black	+2.00
Male: Black or African American	+1.90
Female: Not Hispanic or Latino: Black	+0.80
Female: Black or African American	+0.70
Female: Not Hispanic or Latino: White	+0.60
Male: Hispanic or Latino: All races	+0.50
Female: White	+0.40
Male: Not Hispanic or Latino: Asian or Pacific Islander	+0.20
Female: Hispanic or Latino: All races	+0.20
Male: Asian or Pacific Islander	+0.20
Male: White	+0.00
Male: Not Hispanic or Latino: White	+0.00
Female: Not Hispanic or Latino: Asian or Pacific Islander	-0.10
Female: Asian or Pacific Islander	-0.10
Female: American Indian or Alaska Native	-0.20
Female: Not Hispanic or Latino: American Indian or Alaska Native	-0.30
Male: American Indian or Alaska Native	-1.40
Male: Not Hispanic or Latino: American Indian or Alaska Native	-2.10

Highest change in rate: Male: Not Hispanic or Latino: Black at +2.00

Average change in rate: 0.18

Above avg (10):

- Male: Not Hispanic or Latino: Black	+2.00
- Male: Not Hispanic or Latino: Asian or Pacific Islander	+0.20
- Male: Hispanic or Latino: All races	+0.50
- Female: Not Hispanic or Latino: White	+0.60
- Female: Black or African American	+0.70
- Female: Hispanic or Latino: All races	+0.20

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- Female: Not Hispanic or Latino: Black +0.80
- Female: White +0.40
- Male: Asian or Pacific Islander +0.20
- Male: Black or African American +1.90
Below avg (8):
- Female: Not Hispanic or Latino: Asian or Pacific Islander -0.10
- Male: American Indian or Alaska Native -1.40
- Male: Not Hispanic or Latino: American Indian or Alaska Native -2.10
- Male: White +0.00
- Female: Not Hispanic or Latino: American Indian or Alaska Native -0.30
- Female: American Indian or Alaska Native -0.20
- Male: Not Hispanic or Latino: White +0.00
- Female: Asian or Pacific Islander -0.10

==== Analysis for: Drug overdose deaths involving other synthetic opioids (other than methadone) ====
Race/Ethnicity      Change in rate since 2010
-----
Male: Not Hispanic or Latino: Black +17.30
Male: Black or African American +16.80
Male: Not Hispanic or Latino: White +16.40
Male: White +14.30
Male: Hispanic or Latino: All races +7.30
Female: Not Hispanic or Latino: White +6.30
Female: White +5.20
Female: Not Hispanic or Latino: Black +5.10
Female: Black or African American +4.90
Female: Not Hispanic or Latino: American Indian or Alaska Native +3.40
Female: American Indian or Alaska Native +2.20
Female: Hispanic or Latino: All races +1.60

Highest change in rate: Male: Not Hispanic or Latino: Black at +17.30

Average change in rate: 8.40
Above avg (4):
- Male: Not Hispanic or Latino: White +16.40
- Male: Not Hispanic or Latino: Black +17.30
- Male: White +14.30
- Male: Black or African American +16.80
Below avg (8):
- Female: Not Hispanic or Latino: White +6.30
- Male: Hispanic or Latino: All races +7.30
- Female: Hispanic or Latino: All races +1.60
- Female: White +5.20
- Female: American Indian or Alaska Native +2.20
- Female: Not Hispanic or Latino: American Indian or Alaska Native +3.40
- Female: Not Hispanic or Latino: Black +5.10

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```

- Female: Not Hispanic or Latino: Black +5.10
- Female: Black or African American +4.90

==== Analysis for: All Drugs Combined ====
Race/Ethnicity      Change in rate since 2010
-----
Male: Not Hispanic or Latino: White +6.80
Male: Not Hispanic or Latino: Black +6.60
Male: Black or African American +6.60
Male: White +6.00
Male: Not Hispanic or Latino: American Indian or Alaska Native +4.50
Male: Hispanic or Latino: All races +3.40
Male: American Indian or Alaska Native +2.70
Female: Not Hispanic or Latino: White +2.50
Female: White +2.20
Female: Not Hispanic or Latino: Black +2.10
Female: Black or African American +2.00
Female: Not Hispanic or Latino: American Indian or Alaska Native +1.10
Female: Hispanic or Latino: All races +0.90
Female: American Indian or Alaska Native +0.80
Male: Asian or Pacific Islander +0.10
Male: Not Hispanic or Latino: Asian or Pacific Islander +0.10
Female: Asian or Pacific Islander -0.20
Female: Not Hispanic or Latino: Asian or Pacific Islander -0.20

Highest change in rate: Male: Not Hispanic or Latino: White at +6.80

Average change in rate: 2.67
Above avg (7):
- Male: Not Hispanic or Latino: White +6.80
- Male: Not Hispanic or Latino: American Indian or Alaska Native +4.50
- Male: Hispanic or Latino: All races +3.40
- Male: Not Hispanic or Latino: Black +6.60
- Male: American Indian or Alaska Native +2.70
- Male: Black or African American +6.60
- Male: White +6.00
Below avg (11):
- Male: Asian or Pacific Islander +0.10
- Female: Not Hispanic or Latino: American Indian or Alaska Native +1.10
- Female: Not Hispanic or Latino: Black +2.10
- Female: White +2.20
- Female: Black or African American +2.00
- Male: Not Hispanic or Latino: Asian or Pacific Islander +0.10
- Female: Not Hispanic or Latino: White +2.50
- Female: Hispanic or Latino: All races +0.90
- Female: American Indian or Alaska Native +0.80

```

```

- Female: Asian or Pacific Islander -0.20
- Female: Not Hispanic or Latino: Asian or Pacific Islander -0.20

```

- Interpretation in project context (no need for "groundbreaking" results).

When we look at the data we can see that certain demographic groups have experienced larger absolute increase in overdose death rates since 2010. For all the drugs in the given data we know that the highest change in rate is males not Hispanic or Latino but White at +6.80 for all aspects. And a female Asian or Pacific Islander with a -.02 below the avg. But in certain drugs that involving natural and semisynthetic opioids it would be Black men at +2.00 same goes when we find the avg rate of change. But we did see that in most drugs and in overall drug overdose that Female's who are Asian or Pacific Islander had the smaller changes. In the overall drug useage we see several groups cluster above the overall avg showing the disparities in race and gender. Of course, there are ways to improve the findings by also adding age and geography but also since we considered two time points 2010 and 2018 we could miss non linear trends. Overall using this as a way for community outreach to help these communities.